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21 March 1979
SUPERSEDING
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MILITARY SPECIFICATION

REGULATOR, ENGINE GENERATOR: SOLID STATE, 28 VOLT, 300 AMPERE, DIRECT CURRENT

This specification is approved for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1. Scope. This specification covers a solid state regulator incorporating electronic devices to regulate the voltage, control the reverse current, and limit the current of the 28 volt, 300 ampere (amp), direct current (dc) generator (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2920

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TT-P-1757 - Primer Coating, Zinc Chromate, Low-moisture-sensitivity.

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541 - Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-B-11188 - Battery, Storage, Lead-acid.
MIL-E-13856 - Electrical Components for Automotive Vehicles, Waterproofness Tests.
MIL-G-62061 - Generator, Engine Accessory 28 Volt DC, Rated Output 300 Amps.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1275 - Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.

DRAWINGS

ARMY

11655194 - Process Specification for Soldering, Electrical Connections, for Electrical and Electronic Equipment.
12257823 - DC Solid State Regulator.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. Preproduction samples of the regulator to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The preproduction samples shall be tested and inspected by the contractor under the surveillance of procuring activity personnel to determine conformance to the quality assurance provisions of this specification (see 4.2). The preproduction samples submitted by a contractor shall be fully representative of the regulator to be supplied from production facilities and tooling.

3.1.1 Sample disposition. Regulators subjected to preproduction tests, which are destructive in nature, shall not be delivered for use as defined in 6.1 and shall be indelibly marked: DO NOT USE.

3.2 Materials. Material shall be as specified herein, in applicable specifications, and on applicable drawings. Materials not specifically designated shall be suitable for use in the regulator, operating over an ambient temperature range from minus 65 degrees Fahrenheit (- 65°F) to plus 215°F, without any change in physical or dimensional properties that would result in operation of the regulator falling outside of the limits herein specified (see 3.5.1, 3.5.2 and 6.4).

3.3 Design and construction.

3.3.1 Interchangeability. To provide for interchangeability, the regulator fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of drawing 12257823. Electrical connectors, mounting dimensions and location shall also conform to the detail requirements of drawing 12257823. Contact designation shall be as follows: Contact A is a positive (+) connection to the main engine generator, with battery connection;

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Contact B is unused; Contact C is to the interpole circuit; Contact D is to the generator field; Contacts E and F are unused; Contact G is a positive (+) connection to the main engine generator, with battery connection; and Contact H is to the air cleaner blower relay circuit.

3.3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.3.3 Standard parts. Military standard parts shall be used wherever possible.

3.3.4 Threaded surfaces. All screw threads shall conform to National Bureau of Standards Handbook H28 (see 2.2). The thread form, class and the number of threads-per-inch shall be as specified on the applicable standards or drawings.

3.3.5 Polarity. The regulator shall be designed to operate with a negatively-grounded electrical system, and shall be internally case grounded.

3.3.6 Accessibility. Access to internal components of the regulator shall be attained by removal of the cover. Cover seals shall be provided to prevent entry except by breaking the seals. Installation and removal of the regulator shall be possible without the use of special tools.

3.3.7 Mounting position. The regulator shall be designed to operate when mounted in any position and without the use of shock absorbing material or equipment.

3.3.8 Weight. The regulator weight shall be no more than 16 pounds.

3.4 Electrical.

3.4.1 Reverse current. When tested as specified in 4.5.2.1, the reverse current (I_R) shall not exceed 30 milliamperes when measured between regulator and batteries.

3.4.2 Voltage regulation, preproduction and control. When tested as specified in 4.5.2.2, the regulator shall establish and maintain regulated voltage (E_R) at 28 ± 0.7 volts over a load range of 25 to 350 amperes (see figure 1). The fundamental frequency of the voltage waveform shall be greater than 100 hertz during all portions of the

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test. Throughout the test, the regulator shall re-establish and maintain a regulated voltage within the time and voltage limits indicated on figure 1. Overshoot and undershoot limits shall conform to limits on figure 1. For load currents above 350 amperes with load increased to 400 amperes, the regulated voltage (E_R) shall be equal or less than that measured at 350 amperes.

3.4.2.1 Voltage regulation, quality conformance. When tested as specified in 4.5.2.2.1, the regulated voltage (E_R) shall be maintained at 28 ± 0.7 volts.

3.4.3 Voltage ripple. When tested as specified in 4.5.2.3, the regulator output voltage waveform shall have the following limitations. The ac peak to peak ripple voltage (E_{RV}) shall be no more than 7 volts. With batteries connected, the peak ripple voltage shall be no greater than 4.0 volts, except that excursions of less than 1 milliseconds may exceed 4.0 volts.

3.4.4 Current limit. When tested as specified in 4.5.2.4, the regulator-generator system shall provide the following control:

a. A minimum of 370 amperes within the voltage regulation range of 28 ± 0.7 volts.
b. The regulator shall limit the generator output current within the limits shown in figure 7.
c. The regulator shall attain stable current regulation in the current limit mode within 0.3 second of load application.
d. The regulator shall attain stable voltage regulation upon load release within 0.3 second after entering the lower voltage regulation limit (27.3 volts) shown in figure 8.
e. In the current limit mode, the frequency of oscillation of the dc voltage shall have period of less than 14 milliseconds (Δt-figure 8) and a peak-to-peak value (Δv-figure 8) of less than 10 percent of the dc voltage.

3.4.4.1 Interpole field coil voltage drop, quality conformance. When tested as specified in 4.5.2.4.1, the regulator shall limit the generator interpole coil voltage drop to 1.95 ± 0.05 volts.

3.4.5 Overvoltage. The regulator generator system shall be protected from any failure in the regulator which will cause the regulator to go "full on". When tested as specified in 4.5.2.5, the overvoltage portion of the regulator shall react to deactivate the generator when the system voltage reaches a value of 33.0 ± 1.0 volts. Overvoltage reaction time shall be not less than 0.25 second and not more than 1.00 second. Reset shall be manual and located on the regulator.

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3.4.5.1 Overvoltage, quality conformance. When tested as specified in 4.5.2.5.1, the unstabilized regulator shall have a reaction time of not less than 0.25 second and not more than 1.00 second.

3.4.6 Field excitation. When tested as specified in 4.5.2.6, the regulator shall be activated and begin to regulate when the generator residual voltage increases above 0.7 volts.

3.4.7 Field circuit. When tested as specified in 4.5.2.9, the regulator shall be protected against any inductive energy that may be applied as a result of opening switch S_1 (see figure 3). Five teste cycles shall be conducted and the regulator shall subsequently meet the requirements of 3.4.2.1.

3.4.8 Negative voltage operation. When tested as specified in 4.5.2.10, the regulator shall not be damaged by the application of negative voltage. Subsequent to the test, the regulator shall meet the requirements of 3.4.2.1, 3.4.4.1 and 3.4.5.1.

3.4.8.1 Negative voltage operation, quality conformance. When tested as specified in 4.5.2.10.1, regulator performance shall not be adversely affected by the application of negative voltage.

3.5 Environmental.

3.5.1 High temperature. When tested as specified in 4.5.3.1, the regulator shall meet the requirements of 3.4.1, 3.4.5, 3.4.6 and 3.4.7. The regulator shall also meet the requirements of 3.4.2 except the regulated voltage (E_R) shall be 26.5 $\pm$ 0.7 volts.

3.5.1.1 Transient voltage. While still at $\pm$ 215° $\pm$ 5°F, the regulator shall be tested as specified in 4.5.2.7, and shall withstand the transient environment of MIL-STD-1275.

3.5.1.2 Negative transient voltage. With the ambient temperature maintained at $\pm$ 215° $\pm$ 5°F, the regulator shall be tested as specified in 4.5.2.8 and shall evidence no damage.

3.5.2 Low temperature. When tested as specified in 4.5.3.2, the regulator shall meet the requirements of 3.4.1, 3.4.5, 3.4.6 and 3.4.7. The regulator shall also meet the requirements of 3.4.2 except the regulated voltage (E_R) shall be 29.5 $\pm$ 0.7 volts.

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3.5.3 Vibration. When tested as specified in 4.5.3.3, the regulator shall evidence no breakage, loosening of parts, abrasion, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.4.1, 3.4.2, 3.4.4 and 3.4.5.

3.5.4 Shock. When tested as specified in 4.5.3.4, the regulator shall evidnce no breakage, loosened or distorted parts, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.4.1, 3.4.2, 3.4.4 and 3.4.5.

3.5.5 Corrosion resistance. When tested as specified in 4.5.3.5, the the regulator shall show no evidence of corrosion that would adversely affect performance and shall subsequently meet the requirements of 3.4.2.

3.5.6 Fungus resistance. When tested as specified in 4.5.3.6, the regulator shall not support microbial growth that would adversely affect performance and shall subsequently meet the requirements of 3.4.2.

3.5.7 Waterproofness. When tested as specified in 4.5.3.7, the regulator shall show no evidence of leakage or damage that would adversely affect performance. Subsequent to the test, the regulator shall meet the requirements of 3.4.2.

3.5.8 Electromagnetic interference. When tested as specified in 4.5.3.8, the regulator shall meet the requirements of MIL-STD-461 as defined herein. These requirements shall be met with the regulator operating a balanced 300 amp split load (see figure 5).

3.5.8.1 Radiated emissions. When tested as specified in 4.5.3.8, at a one meter distance, the broadband (BB) and narrowband (NB) radiated emissions in the frequency range of 150 KHz to 10 GHz shall not exceed the limits of MIL-STD-461 subtest RE02 (BB) and RE02.1 (NB).

3.5.8.2 Conducted emissions. When tested as specified in 4.5.3.8, on a + 28 volt dc, 50 amp lead of the split load, the broadband and narrowband conducted emissions in the frequency range of 50 KHz to 50 MHz shall not exceed the limits of MIL-STD-461 subtest CE04.

3.6 Endurance. The regulator shall be tested for 1000 hours as specified in 4.5.4. There shall be no servicing, adjustment or replacement of parts throughout the test. Subsequent to the test, the regulator shall meet the requirements of 3.4.1 through 3.4.6 and 3.5.7.

3.7 Finish. Unless otherwise specified (see 6.2), finish shall be as follows:

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3.7.1 Painting. All exterior surfaces of the regulator assembly shall be painted, except the electrical connections and mating surfaces on the mounting bracket.

3.7.2 Ferrous surfaces. Ferrous metal surfaces shall be cleaned in accordance with applicable method specified in TT-C-490. The cleaned surfaces shall be treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 to 1.25 mil thick or TT-P-1757, 0.40 to 0.60 mil thick. The finish coat shall white gloss enamel conforming to TT-E-469, 0.75 to 1.25 mil thick.

3.7.3 Aluminum alloy surfaces. Aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541 or anodized in accordance with MIL-A-8625. Exterior surfaces shall be given one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil thick. The finish coat shall be white gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.7.4 Zinc and cadmium surfaces. Exterior zinc and cadmium surfaces shall be given one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil thick. The finish coat shall be white gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.8 Identification.

3.8.1 Marking. The regulators shall be marked in accordance with MIL-STD-130.

3.8.2 Nameplate. A nameplate conforming to the nondimensional requirements of MIL-P-514 shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the plate shall include the following as a minimum:

Regulator
28 volts dc
Manufacturer's identification
Serial number
Military part number
Federal stock number
Contract number
Date of manufacture
US

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3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connections, cracked insulation, or other defects which affect serviceability or appearance. Soldering must conform to requirements specified in drawing 11655194.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. The preproduction sample (see 3.1) shall consist of four regulators. The regulators, properly marked with identifying information, shall be representative of the regulators proposed to be furnished to the Government. Unless otherwise specified, preproduction testing shall be conducted by the contractor at a place approved by the procuring activity and shall consist of examination for defects specified in 4.3.2.2, and testing as specified in table I in the order listed. The procuring activity shall be notified prior to initiation of preproduction inspection.

NOTE: Regulators subjected to preproduction tests shall not be used for any other purpose and shall be indelibly marked: DO NOT USE (see 3.1.1).

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TABLE I. Order of preproduction testing.

Test	Paragraph no.	Specimens 1	2	3	4
Reverse current	4.5.2.1	X	X	X	X
Voltage regulation	4.5.2.2	X	X	X	X
Voltage ripple	4.5.2.3	X	X	X	
Current limit	4.5.2.4	X	X	X	
Field excitation	4.5.2.6	X	X	X	
High temperature (/ 215°- / 5°F)	4.5.3.1		X	X	
Low temperature (-65°/ 5°F) -	4.5.3.2		X	X	
Endurance	4.5.4	X			
Vibration	4.5.3.3		X	X	
Shock	4.5.3.4		X	X	
Waterproofness	4.5.3.7.1	X	X	X	
Corrosion resistance	4.5.3.5		X		
Transient voltage	4.5.2.7		X	X	
Overvoltage	4.5.2.5	X	X	X	
Fungus resistance	4.5.3.6			X	
Electromagnetic interference-	4.5.3.8				X
Field circuit	4.5.2.9	X	X	X	
Negative voltage operation	4.5.2.10	X	X		

4.2.1 Failure. Failure of a preproduction specimen to pass any specified examination or test may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all regulators from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.2 Examination.

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4.3.2.1 Acceptable quality level. Each regulator selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table II.

TABLE II. Classification of defects.

Categories	Defects	Method examination
Major:		
101 -----	Incorrect dimensions affecting interchangeability (see 3.3.1)	Scale
Minor:		
201 ------	Incorrect dimensions not affecting interchangeability (see 3.3.1)	Scale
202 ------	Improper finish (see 3.7)	Visual
203 ------	Improper marking (see 3.8)	Visual
204 ------	Faulty workmanship (see 3.9)	Visual

4.3.3 Testing.

4.3.3.1 Quality conformance. All regulators, selected as specified in 4.3.1.2, shall be subjected to the tests listed in table III in the order listed.

TABLE III. Quality conformance tests.

Paragraph no.	Test
4.5.2.2.1	Voltage regulation
4.5.2.4.1	Interpole field coil voltage drop
4.5.2.5.1	Overvoltage
4.5.2.6	Field excitation
4.5.2.10.1	Negative voltage operation
4.5.3.7.2	Waterproofness

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4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of three (3) for each 500 units or less produced, except that not less than three nor more than six units shall be selected from a lot which has passed the examination specified in 4.3.2 and the tests specified in table III, and shall be subjected to the tests in table IV in the order listed.

TABLE IV. Control tests.

Test	Paragraph no.	Specimens 1	2	3
Reverse current	4.5.2.1	X	X	X
Voltage regulation	4.5.2.2	X	X	X
Voltage ripple	4.5.2.3	X	X	
Current limit	4.5.2.4	X	X	X
Overvoltage	4.5.2.5	X	X	X
Field excitation	4.5.2.6	X	X	X
High temperature (+ 215° + 5°F)	4.5.3.1		X	X
Low temperature (-65° + 5°F)	4.5.3.2		X	X
Vibration 1/	4.5.3.3		X	X
Shock 1/	4.5.3.4		X	X
Waterproofness	4.5.3.7.1	X	X	X
Field circuit	4.5.2.9	X	X	X
Negative voltage operation 1/	4.5.2.10	X	X	

1/ Only two specimens shall be subjected to the shock, vibration and negative voltage operation test from each procurement lot of 500 or less.

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause, at the option of the Government, to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, tests shall be conducted at an ambient room temperature of

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80° ± 15°F, and the test circuit (see figure 3) shall include the regulator, mounted on a horizontal plane, battery and generator. The generator used in the test circuit shall conform to MIL-G-62061. The batteries shall consist of six 12-volt batteries conforming to type 6 TN of MIL-B-11188. The six batteries shall be connected in series-parallel. Batteries shall be fully charged (see 6.3.2) and kept at room temperature.

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table V, VI and applicable test specification), test apparatus shall include chambers for conducting low temperature (-80°F), high temperature (± 215°F) tests and a submersion tank.

4.5.1.3 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance, except as indicated in tables V and VI. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to authorize an inspection where accuracy of test equipment has not been established to the satisfaction of the Government.

TABLE V. Measuring instrument tolerances for preproduction tests.

Measuring instrument	Accuracy
Voltmeter dc	± 0.35 percent (40 volts-full scale)
Ammeter dc	± 0.50 percent (600 amps-full scale)
Tachometer	± 1 percent
Milliammeter	± 0.50 percent (600 milliamps-full scale)
Oscilloscope	± 3 percent

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TABLE VI. Measuring instrument tolerances for quality conformance and control tests.

Measuring instrument	Accuracy
Voltmeter dc	± 0.15 volts
Ammeter dc	± 3.0 amps
Tachometer	± 50 rpm at 1,900 rpm
	± 200 rpm at 10,000 rpm
Milliammeter	± 5.0 milliamps
Oscilloscope	± 3 percent

4.5.2 Electrical.

4.5.2.1 Reverse current. To determine conformance to 3.4.1, the temperature stabilized regulator (see 6.3.1) shall be connected in the test circuit (see figure 3) with the generator operating at sufficient speed to deliver charging current to the batteries. The generator speed shall then be gradually reduced to zero. The maximum values of the reverse current (I_R) shall be observed on the milliammeter. Five test cycles shall be recorded.

4.5.2.2 Voltage regulation, preproduction and control. To determine conformance to 3.4.2, the temperature stabilized regulator shall be connected in the test circuit. During the test, the resistance load shall be applied and removed suddenly. Change in load shall be accomplished in 10 milliseconds maximum time. Tripping of overvoltage reset due to quick reaction shall not constitute failure. Observation shall be made of the output voltage while the regulator is being subjected to the following test procedure:

	RPM	Load (AMPS)
a.	2400 ± 50	25
	2400 ± 50	100
	2400 ± 50	350
b.	5500 ± 110	same sequence as in (a)
c.	8000 ± 160	same sequence as in (a)

E_R to be recorded on CEC tapes, or equivalent.

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4.5.2.2.1 Voltage regulation, quality conformance. To determine conformance to 3.4.2.1, the generator shall be operated at 5500 rpm with a 350 ampere load for five minutes. The regulated voltage (E_R) shall then be recorded with the unstabilized regulator at an ambient temperature of $80^\circ \pm 15^\circ$F under the following conditions:

RPM	Load (AMPS)
2400 ± 50	Battery load only
2400 ± 50	350
8000 ± 160	Battery load only
8000 ± 160	350

4.5.2.3 Voltage ripple. To determine conformance to 3.4.3, using the test procedure specified in 4.5.2.2, the output voltage waveform at the regulator output shall be measured with an oscilloscope (see figure 3). Repeat the procedure with batteries disconnected.

4.5.2.4 Current limit. To determine conformance to 3.4.4, the regulator-generator system shall be operated, with no batteries connected, at 5500 rpm; with a resistive load rated 350 amps at 28 volts (nominal). Generator inlet cooling air shall be ducted. Ducting configuration shall simulate that used on the M60A1 vehicle generator installation. Temperature shall be measured at a distance of 3 feet 6 inches from the fan centerline. The regulator-generator shall be operated until circuit conditions are stabilized. The following tests shall then be performed (under static circuit conditions, using an average reading meter for voltage and current readings) (for transient circuit conditions, appropriate instrumentation shall be used):

a. To determine conformance to 3.4.4 a, the load resistance shall be reduced in steps not to exceed 6 ohms.
b. To determine conformance to 3.4.4 b, c, d and e, a resistive load bank of 1000 amps capacity at 28 volts (nominal) shall be suddenly applied and released.

4.5.2.4.1 Interpole field coil voltage drop, quality conformance. To determine conformance to 3.4.4.1, the batteries shall be disconnected and the regulator-generator system operated at 5500 rpm with a 350 amp resistive load. Switch the output into a resistive load bank of 1000 amps at 28 volts and measure the voltage drop across the interpole field coil at the generator terminals.

4.5.2.5 Overvoltage. To determine conformance to 3.4.5, the regulator shall be connected to a dc power supply. The voltage shall be increased from 28 volts by application of a step voltage of approximately 6 volts. The voltage level and reaction time shall be measured and recorded.

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4.5.2.5.1 Overvoltage, quality conformance. To determine conformance to 3.4.5.1, the unstabilized regulator shall be disconnected from the generator and a dc voltage of 6 volts shall be applied upon the basic 28 volts. The voltage level and reaction time shall be measured and recorded.

4.5.2.6 Field excitation. To determine conformance to 3.4.6, the regulator shall be connected in the circuit (see figure 3) with batteries disconnected and the generator speed gradually increased from zero until field excitation occurs. The generator voltage at excitation shall be recorded. The generator speed at start of regulation shall be recorded.

4.5.2.7 Transient voltage. To determine conformance to 3.5.1.1, the regulator shall be disconnected from the test circuit (see figure 3) and stored for 24 hours at 215° ± 5°F. The test shall then be conducted in an ambient temperature of 215° ± 5°F. A voltage pulse of 100 volts, square wave, not more than 160 microseconds rise time, 50 milliseconds duration shall be applied to the regulator output terminal once every six seconds equally spaced for a period of three hours. The test shall be repeated except that a voltage pulse of 70 volts shall be applied to the input terminal of the regulator.

4.5.2.8 Negative transient voltage. To determine conformance to 3.5.1.2, the regulator shall be temperature stabilized at 215° ± 5°F for a period of one hour before being subjected to five cycles of the following two step test procedure, using the test circuit shown in figure 2. Steps one and two shall constitute a cycle and the cycle shall be repeated five times.

Step 1:
a. Capacitor C shall be charged to a voltage (E_C) of 595 ± 5 volts
b. Switch S_2 shall be closed (E_C is applied to the input terminal of the regulator) for a period of six seconds and then opened.

Step 2:
a. With the generator running at a speed between 2400 and 3000 rpm, switch S_3 shall be closed for six minutes.
b. Capacitor C shall be charged to an E_C of 595 ± 5 volts and switch S2 closed for a period.
c. Open all switches.

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4.5.2.9 Field circuit. To determine conformance to 3.4.7, the regulator shall be tested as follows:

a. With the generator operating 2400 rpm in the circuit in figure 3 with a 30 ampere load, switch S_1 shall be opened and closed five times. The load shall be increased to 400 amperes and the procedure repeated.
b. Procedure (a) shall be repeated with the generator operating at 8000 rpm.

Subsequently the regulator shall meet the requirements of 3.4.2.1.

4.5.2.10 Negative voltage operation. To determine conformance to 3.4.8, the regulator shall be operated until temperature stabilization. Then it shall be subjected to the following test procedure using the test circuit shown in figure 6 (batteries shall be fully charged and in satisfactory condition as specified in 4.5.1.1):

a. Arrange the test set-up as shown on the circuit diagram, with S_1 , S2 , and S_3 open.
b. Close S_2 for one to five (1-5) seconds, then open S_2 .
c. Close S_3 with generator at 2400 ± 50 rpm. Observe generator output (IG), which should be less than 20 amps. If IG indicates 200 amps or more, open S_3 and discontinue test. If generator output (IG) does not exceed 20 amps, close S_1 and test regulator in accordance with 4.5.2.2.1, 4.5.2.4.1 and 4.5.2.5.1.
d. Open S_1 , S_2 , and S_3 . Lock generator to prevent rotation (rotation less than 10 rpm is permissible). Close S_3 and reverse the polarity of V_2 for 10 seconds, then return polarity to normal. Repeat step (c) and determine that output is within specification.

NOTE: Steps (b) and (c) test for ability to withstand reverse generator operation. Step (d) tests the ability to withstand reverse slave battery operation.

4.5.2.10.1 Negative voltage operation, quality conformance. To determine conformance to 3.4.8.1, the regulator shall be subjected to the following test procedure using the test circuit of figure 6:

a. Close S_2 from 1 to 5 seconds, then open.
b. Lock the generator to prevent rotation (rotation of less than 10 rpm is permissible). Close S_3 and reverse the polarity of V2 for 10 seconds. Return polarity to normal.

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NOTE: Step (a) tests for ability to withstand reverse generator operation. Step (b) tests for ability to withstand reverse slave battery operation.

4.5.3.1 High temperature. To determine conformance to 3.5.1, the regulator shall be stored for 24 hours in an ambient temperature of 215° ± 5°F. With ambient temperature maintained at 215° ± 5°F, the regulator shall function continuously for four hours while the generator speed and output load are adjusted to produce 2400 rpm and 400 amperes. While remaining in an ambient temperature of 215° ± 5°F the regulator shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.5, 4.5.2.6, 4.5.2.7, 4.5.2.8 and 4.5.2.9. Regulator shall be returned to an ambient temperature of 80° ± 5°F and tested as specified in 4.5.2.2.1, 4.5.2.4.1 and 4.5.2.5.1.

4.5.3.2 Low temperature. To determine conformance to 3.5.2, the regulator shall be stored for 24 hours in an ambient temperature of minus 80° ± 5°F. After stabilizing for four hours at minus 65° ± 5°F, the regulator shall function continuously for one hour while the generator speed and output load are adjusted to 2400 rpm and 400 amperes. While remaining in an ambient temperature of minus 65° ± 5°F, the regulator shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.5, 4.5.2.6 and 4.5.2.9. The regulator shall then be returned to an ambient temperature of plus 80° ± 5°F and tested as specified in 4.5.2.1, 4.5.2.2.1, 4.5.2.4.1 and 4.5.2.5.1.

4.5.3.3 Vibration. To determine conformance to 3.5.3, the regulator, mounted as in intended use, shall be subjected to tests in accordance with MIL-STD-810, method 514, procedure VII, curve W, figure 5, parts 1, 2 and 3, except the loading shall be 12 Gs. The regulator shall subsequently be tested in accordance with 4.5.2.2.1, 4.5.2.4.1 and 4.5.2.5.1. During vibration testing, the relay shall be energized and monitored.

4.5.3.4 Shock. To determine conformance to 3.5.4, the regulator shall be subjected to tests in accordance with MIL-STD-810, method 516, under 40G shock loads of 18 milliseconds duration per procedure I for ground equipment, figure 516-1. The regulator shall subsequently be tested in accordance with 4.5.2.1, 4.5.2.2.1, 4.5.2.4.1 and 4.5.2.5.1. During shock testing, the relay shall be energized and monitored.

4.5.3.5 Corrosion resistance. To determine conformance to 3.5.5, the regulator shall be subjected to the corrosion resistance test specified in method 509 of MIL-STD-810. Regulators shall be tested for four test cycles (192 hours), sealed, and internal components shall not be exposed. The regulator shall subsequently be tested in accordance with 4.5.2.2.

4.5.3.6 Fungus resistance. To determine conformance to 3.5.6, documentation shall be furnished providing evidence that exposed materials of the regulator are non-nutrient to, or have been treated

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to, resist fungi detrimental to the performance of the regulator. In the absence of documentation, the regulator shall be subjected to the fungus resistance test in accordance with method 508, MIL-STD-810, except that the period of exposure shall be 90 days. Sealed and internal components shall not be exposed. The regulator shall subsequently be tested in accordance with 4.5.2.2.1.

4.5.3.7 Waterproofness. The regulator shall be subjected to waterproofness tests in accordance with 4.5.3.7.1 or 4.5.3.7.2.

4.5.3.7.1 Preproduction and control. For preproduction and control test purposes, regulators shall be subjected to the waterproofness test specified in MIL-E-13856 for type II, class 2 components to determine conformance to 3.5.7, except 10 psi pressure and 10 psi vacuum shall be used. The regulator while submerged shall be operable and tested in accordance with 4.5.2.2.1 during the pressure and vacuum tests phases. The regulator shall subsequently be tested in accordance with 4.5.2.2.1 and 4.5.2.4.1.

4.5.3.7.2 Quality conformance. For quality conformance test purposes, non-operating regulators shall be subjected to the quality conformance test inspection specified in MIL-E-13856 for type II, class 2 components to determine conformance to 3.5.7. During submersion, the regulator shall be subjected to 10 psig for a period of five minutes.

4.5.3.8 Electromagnetic interference. To determine conformance to 3.5.8, 3.5.8.1 and 3.5.8.2, the regulator shall be laboratory tested in accordance with MIL-STD-462 and figure 5.

4.5.4 Endurance. To determine conformance to 3.6, the regulator shall be temperature stabilized (see 6.3.1). The regulator shall be cycled at 18 cycles per hour for 1000 hours. The 1000 hours shall be divided into 500 hours with generator operating at 2400 rpm, 300 hours at 4000 rpm, 150 hours at 6000 rpm and 50 hours at 8000 rpm. A cycle shall consist of generator operation for 25 seconds each at battery load, 100 amps, 200 amps, 300 amps and 400 amps, followed by a reduction to 300 amps, 200 amps, 100 amps and then battery load. Subsequently the regulator shall be tested as specified in 4.5.2.1 through 4.5.2.6 and 4.5.3.7.

4.6 Inspection of packaging.

4.6.1 Materials and processes. All materials and processes involved in packaging shall be inspected to determine conformance to requirements of section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The regulators covered by this specification are intended for use in 28 volt dc nominal electrical systems equipped with charging generators. The regulators are intended for use on tactical and combat military vehicles and industrial applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Military part number.
c. Treatment and painting if other than specified (see 3.7).
d. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Definitions.

6.3.1 Temperature stabilized regulator. A temperature stabilized regulator is defined as a regulator which has been operated for 30 minutes with the generator speed adjusted to 2400 rpm, the load adjusted to 175 amps and the temperature at 80° ± 15°F (see 4.5.2.1).

6.3.2 Fully charged battery. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than four amps at 28 volts at room temperature, 80° ± 15°F (see 4.5.1.1).

6.3.3 Load current. Load current, for purposes of this specification, shall be considered as that current from generator and regulator to batteries and resistive load (see figure 3).

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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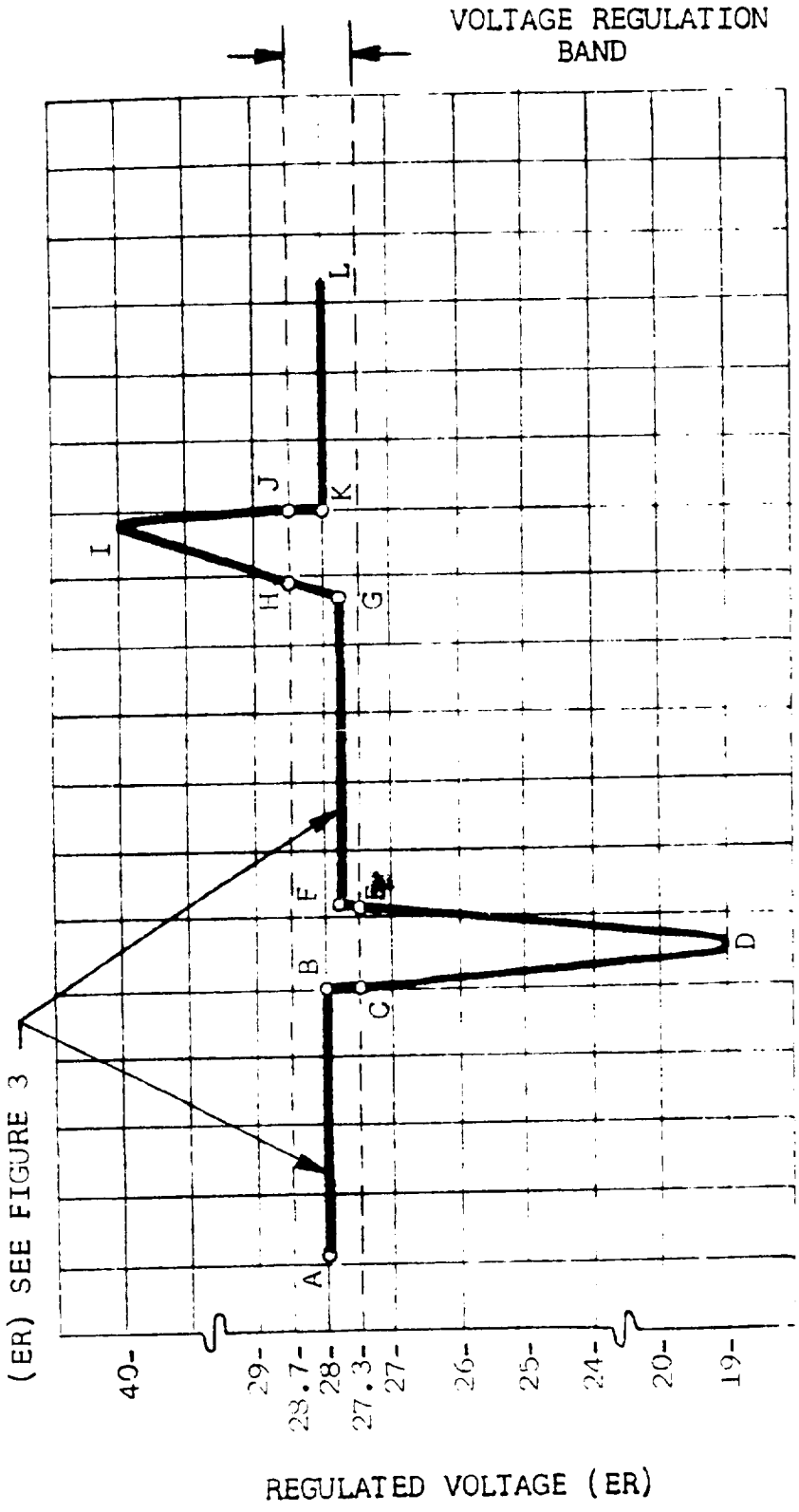
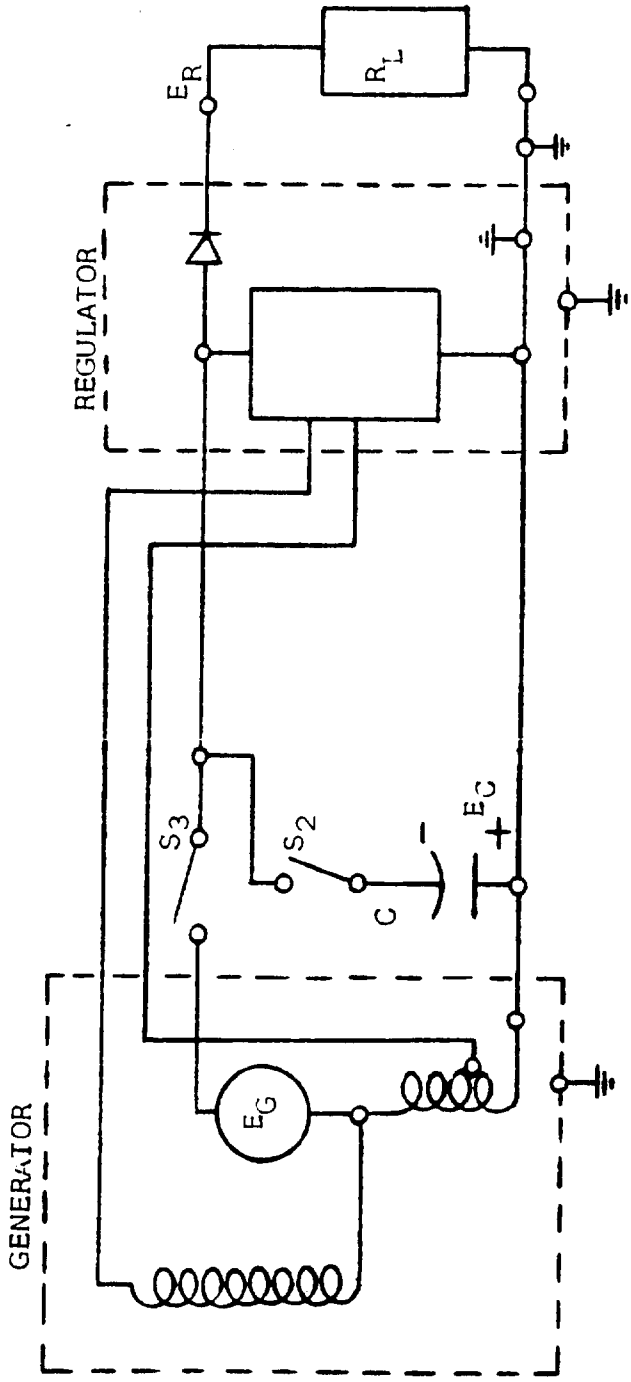


FIGURE 1. Voltage regulation limits.

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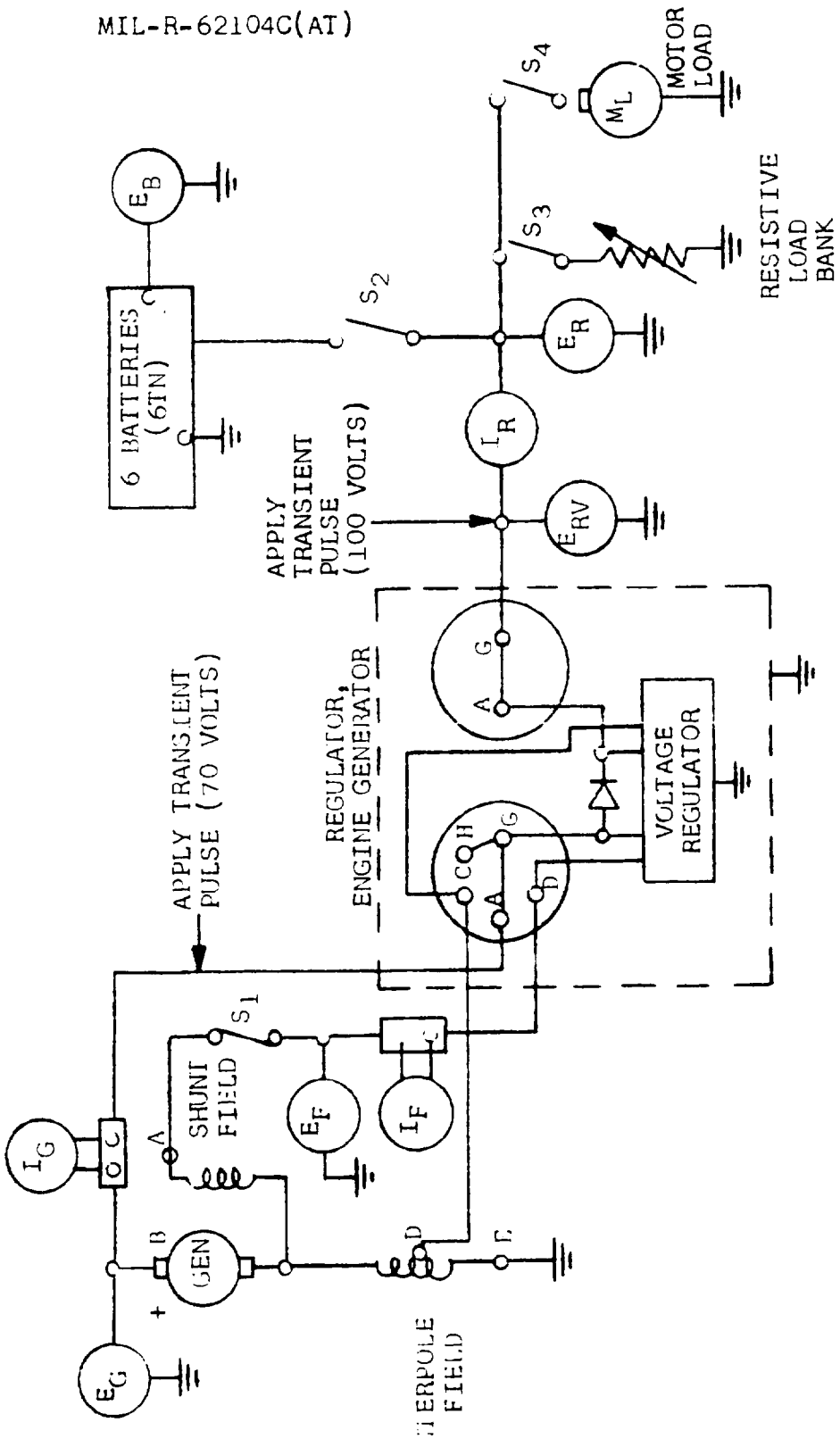


E_C = 595 ± 5 volts
E_R = Regulated voltage

C = 5.3 ± .1 ufd
R_L = 50 ± 5 amps load at 28 volts

FIGURE 2. Test circuit.

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TEST DATA REQUIRED

I_G - GEN AMPS
E_G - GEN VOLTS
E_F - GEN FIELD VOLTS
I_F - GEN FIELD AMPS

E_B - BATTERY VOLTS
E_R - REGULATED VOLTAGE
E_{RV} - RIPPLE VOLTAGE
I_R - REVERSE CURRENT

S_1 - SWITCH USED DURING FORDING MS39061-1
R_L - RESISTIVE LOAD BANK
M_L - MOTOR LOAD (SEE FIG. 4)

FIGURE 3. Test circuit.

(See M60 comments)

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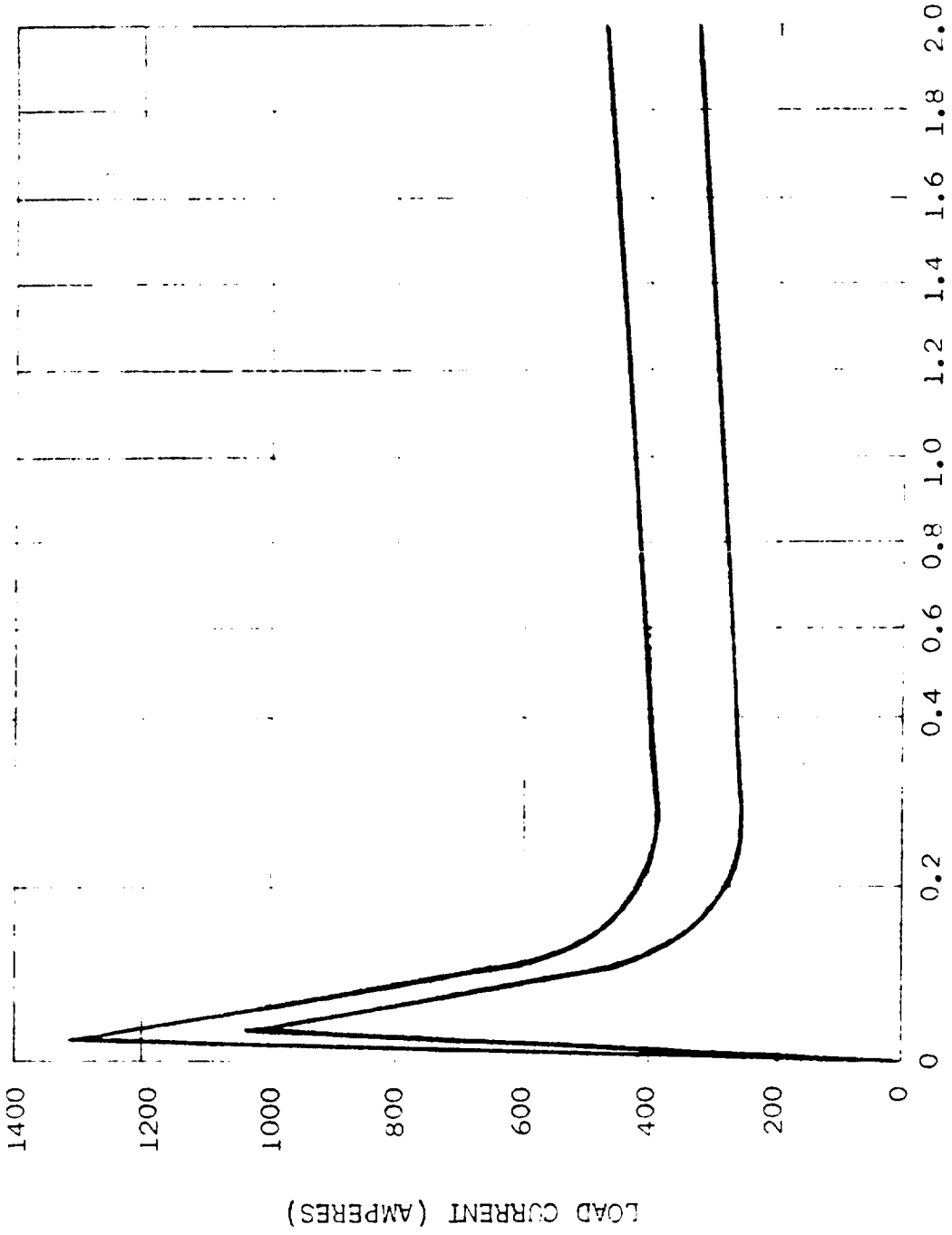


FIGURE 4. Motor load.

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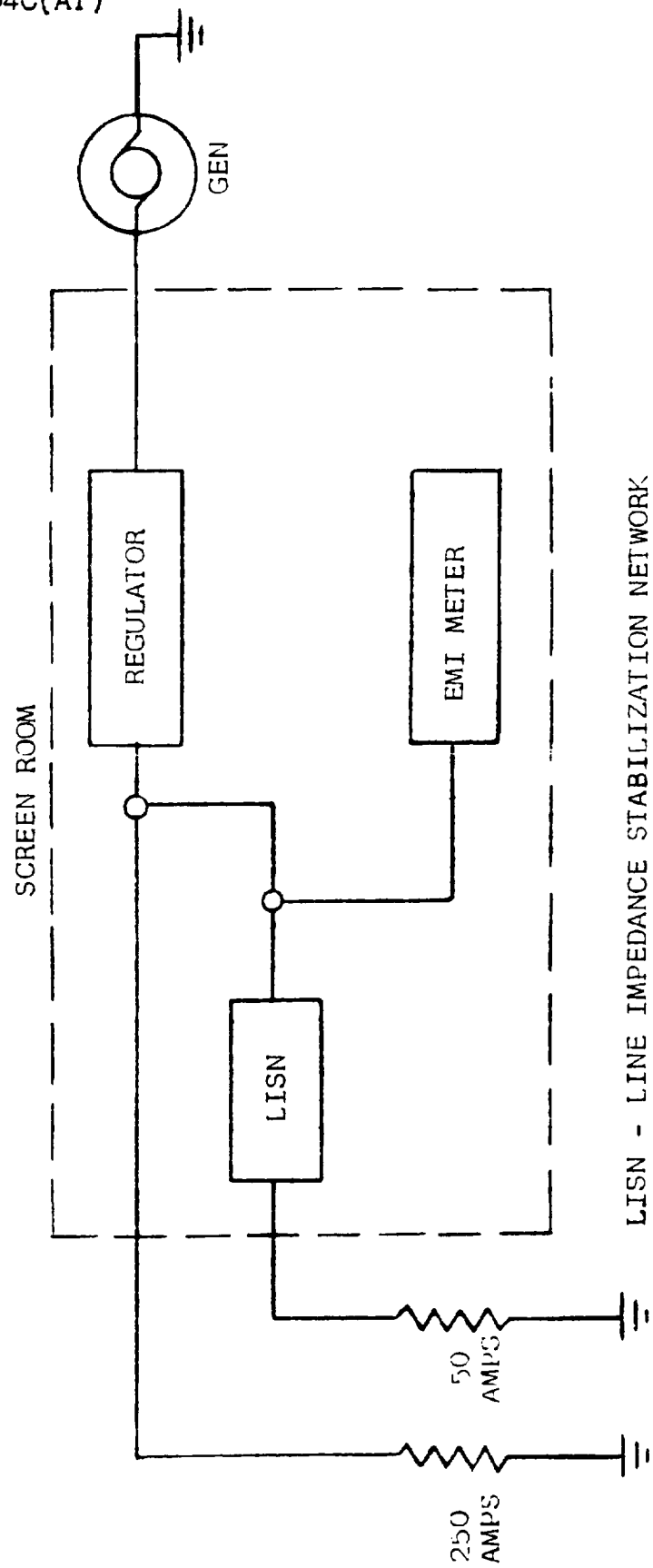
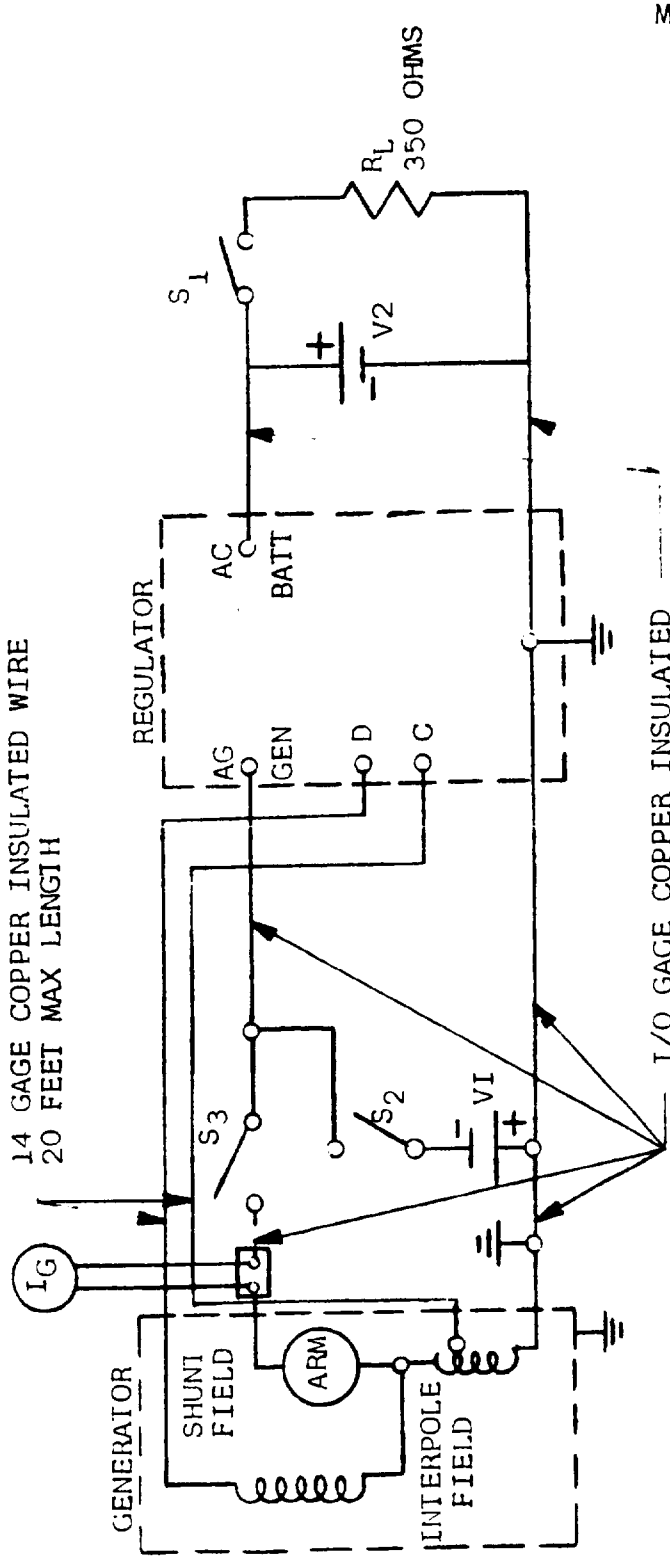


FIGURE 5. Electromagnetic interference tests.

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$V_1 = V_2 = 24$ VDC (TWO 12V - 6TN BATTERIES, OR EQUIVALENT)

FIGURE 6. Negative voltage circuit.

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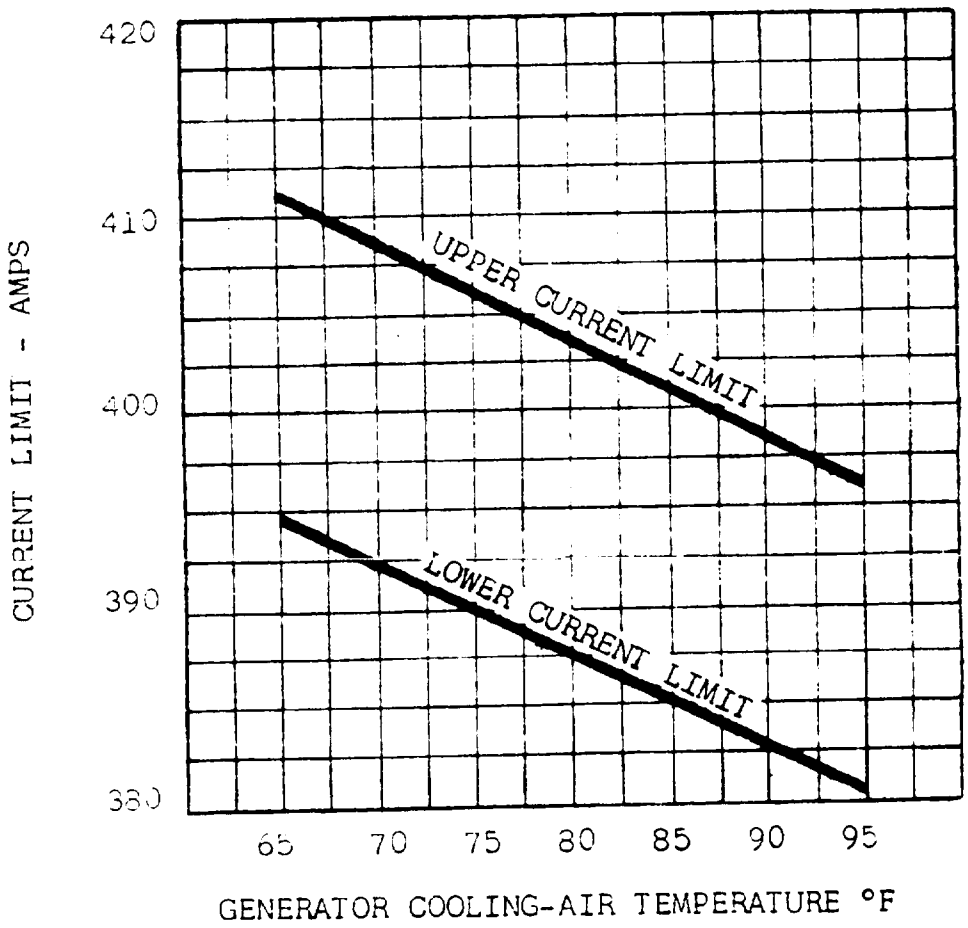


FIGURE 7. Current limit vs cooling air temperature.

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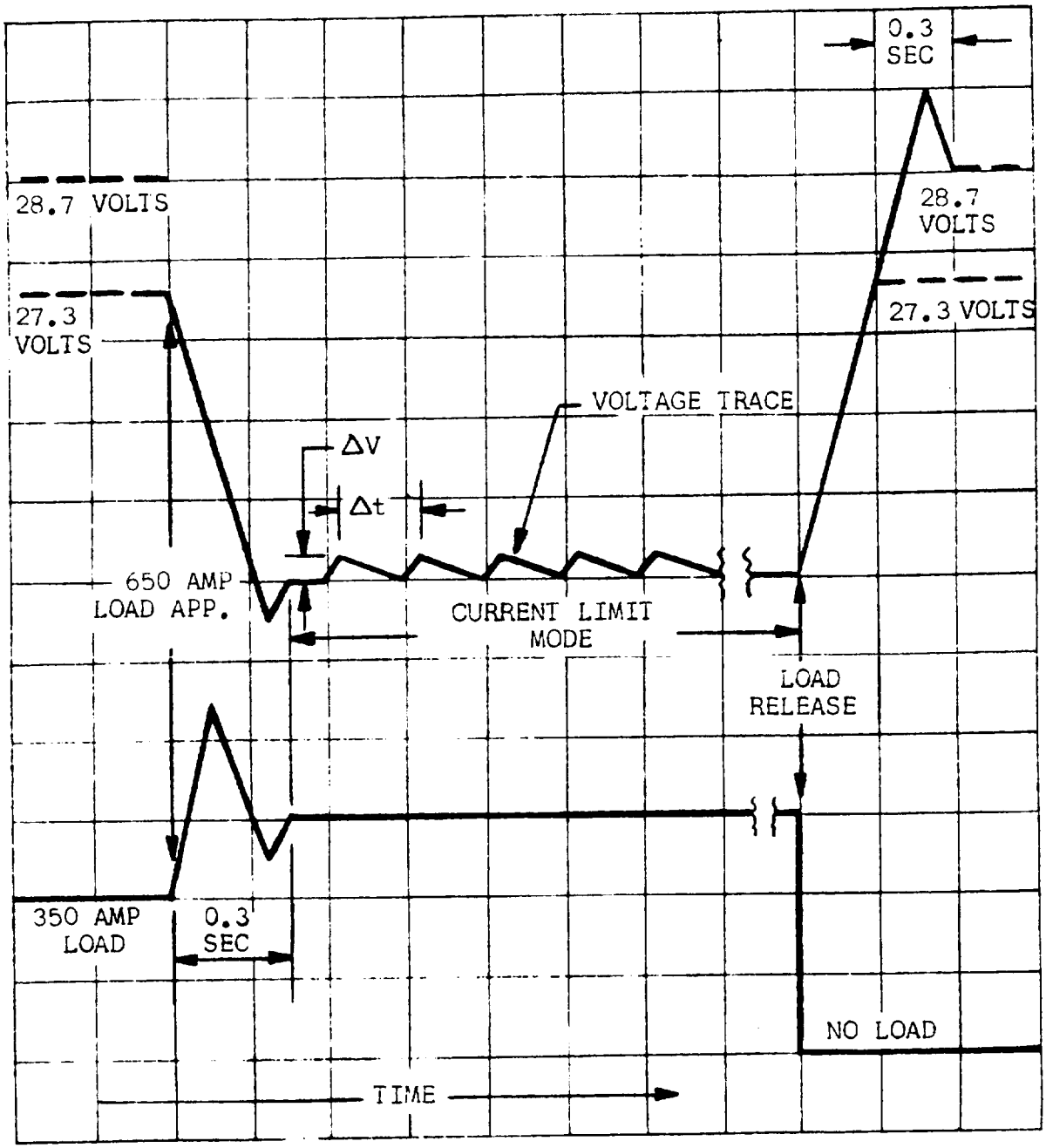


FIGURE 8. Current and voltage recovery time.

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